

TayTech[®] | UK



INDIRECT SMART HEXA SH

Indirect Smart Hexa SH

1. General Description



SmartHexa SH has electronic controller in order to maintain maximum efficiency and has the priority of generating hot water.

SmartHexa SH measures temperatures and pressure values from many different points throughout the unit.

Material of heat exchanger and pipes are stainless steel and HIU can work efficiently in conjunction with aluminum radiators.

An outdoor compensation control can be added optionally in central heating system installed buildings.

The SmartHexa use of the latest electronic stepper motor technology that allow the unit to get stable SH for end users.

The electronic controls also enable to carry intelligent functions, such as PC connectivity for set-up and commissioning, district return temperature control. The SmartHexa can be send all data to BLES system and can be monitored by end users.

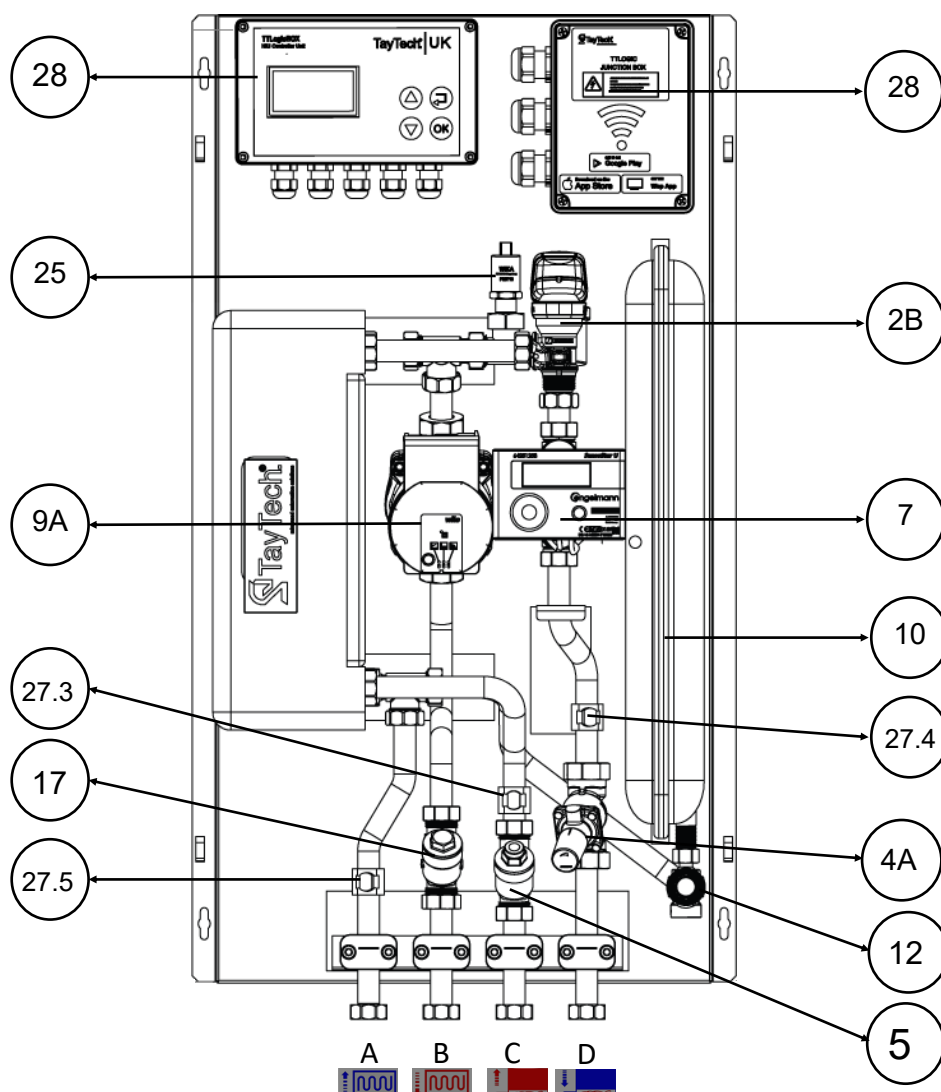
Features

- Fully electronic controlled with PC connectivity via GSM or Ethernet for set-up and commissioning
- Fast SH Controls with calibrated sensors
- Return temperature control for district heating with high efficiency heat exchanger
- Pump and Valve Exercise mode for anti-jamming

Indirect Smart Hexa SH

2. Hydraulic Review

2.1 Component List

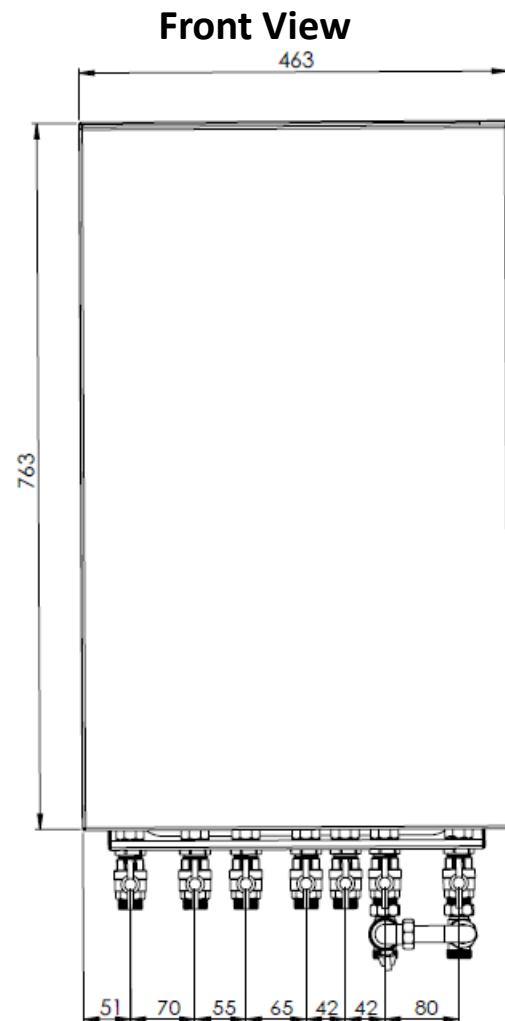
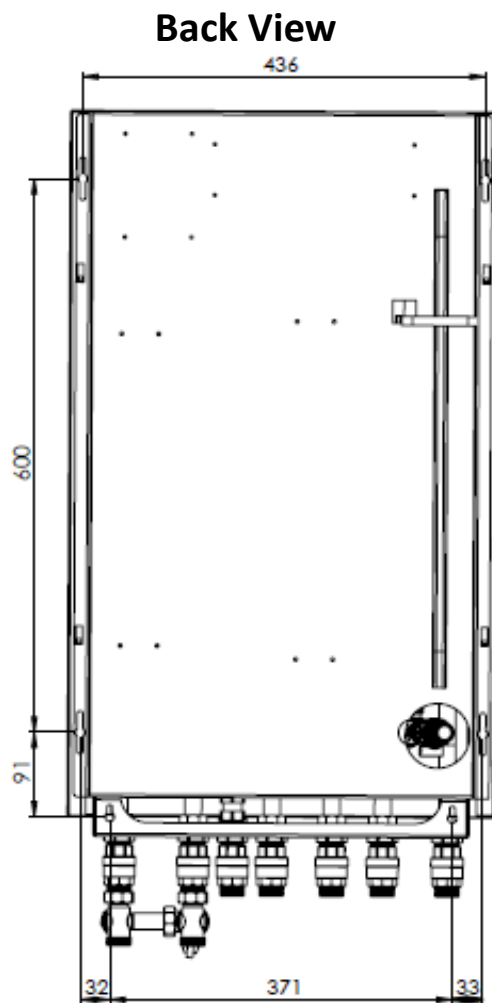


1B SH Heat Exchanger
 2B SH Water Temperature Control Valve
 4A Differential Pressure Valve
 5 District Heating Strainer
 7 Heat Meter
 9B Space Heating Circulation Pump
 10 Expansion Vessel
 11 SH Strainer
 12 Safety Valve
 24 Bleed Point (Automatic)
 26 Junction Box
 25 Space Heating Pressure Sensor
 27.3 District Heating Temperature Sensor
 27.4 SH Return Temperature Sensor
 27.5 SH Temperature Sensor
 28 TT LogicBOX Controller

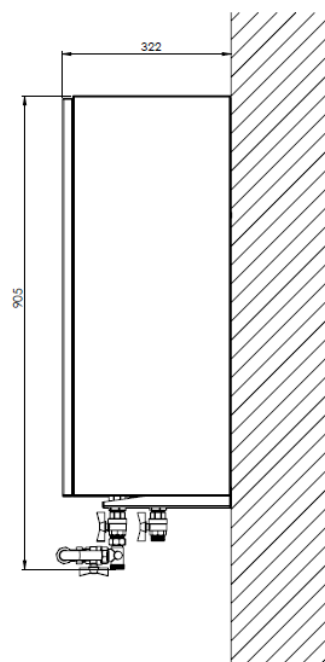
A	-	Space Heating Supply
B	-	Space Heating Return
C	-	District Heating Supply
D	-	District Heating Return

Indirect Smart Hexa SH

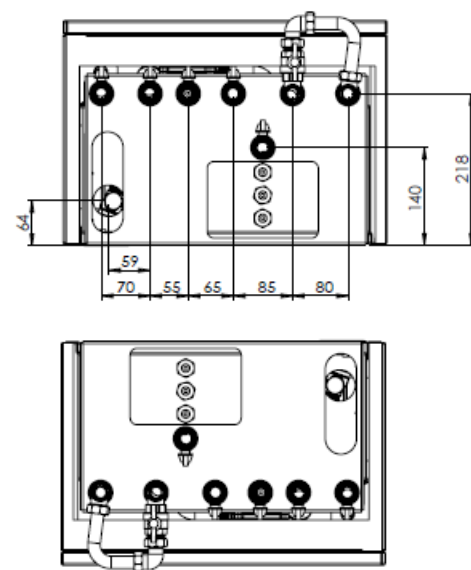
2. Dimensions



Wall Mounted View



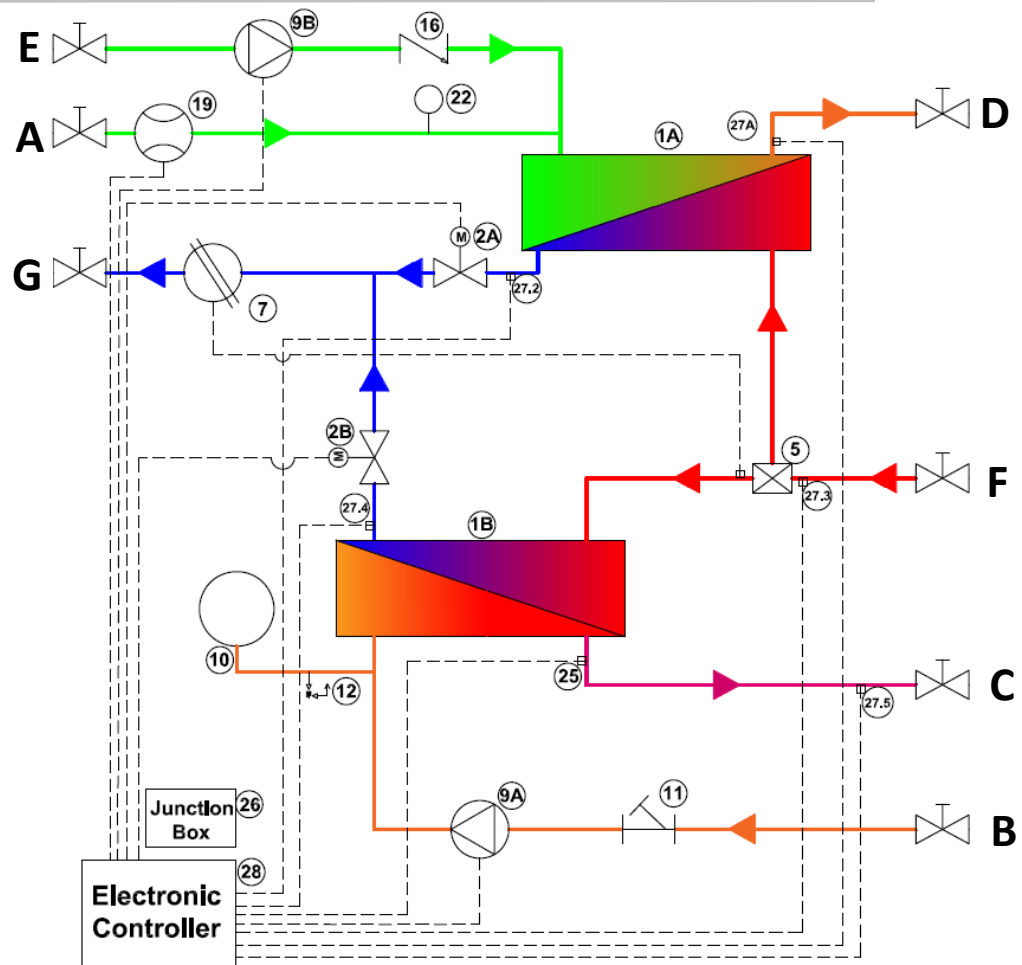
Bottom View



Indirect Smart Hexa SH

2. Hydraulic Review

2.2 Flow Diagram



A —	DCWI	—	Domestic Cold Water Inlet
B —	SHR	—	Space Heating Return
C —	SHS	—	Space Heating Supply
D —	DHW	—	Domestic Hot Water
E —	Re-Circ	—	Re-Circulation
F —	DHS	—	District Heating Supply
G —	DHR	—	District Heating Return

★ Indirect Smart Hexa SH

3. Technical Specifications

Description	Data
Nominal primary supply temperature	80°C
Maximum primary supply temperature	85°C
Minimum primary supply temperature	55°C
Return Temp Limit Range	35°C to 65°C
CH Limit Range	30°C to 80°C
Primary connections	Female / 18mm
Mains & DHW connections	Female / 18mm
Central heating connections	Female / 18mm
Pressure relief	15mm
Maximal primary differential pressure (without dP Valve)	450 kPa
Pressure class DH circuit	PN16
Pressure class CH circuit (3 bar safety valve)	PN10
Pressure class DHW circuit	PN10
SH Maximum working pressure	2.5 Bar
Casing width	450 mm
Casing height	787 mm
Casing depth	220 mm
Casing Material	Metal case with Insulation
Weight	25kg **
Electrical supply info	230V 50Hz
Fuse ratings	3 Amp
Sensor DHW	NTC 10k ohm @ 25°C
Functions	Pump Exercise Valve Exercise Pre-pay Shut Off Primary Return Temperature

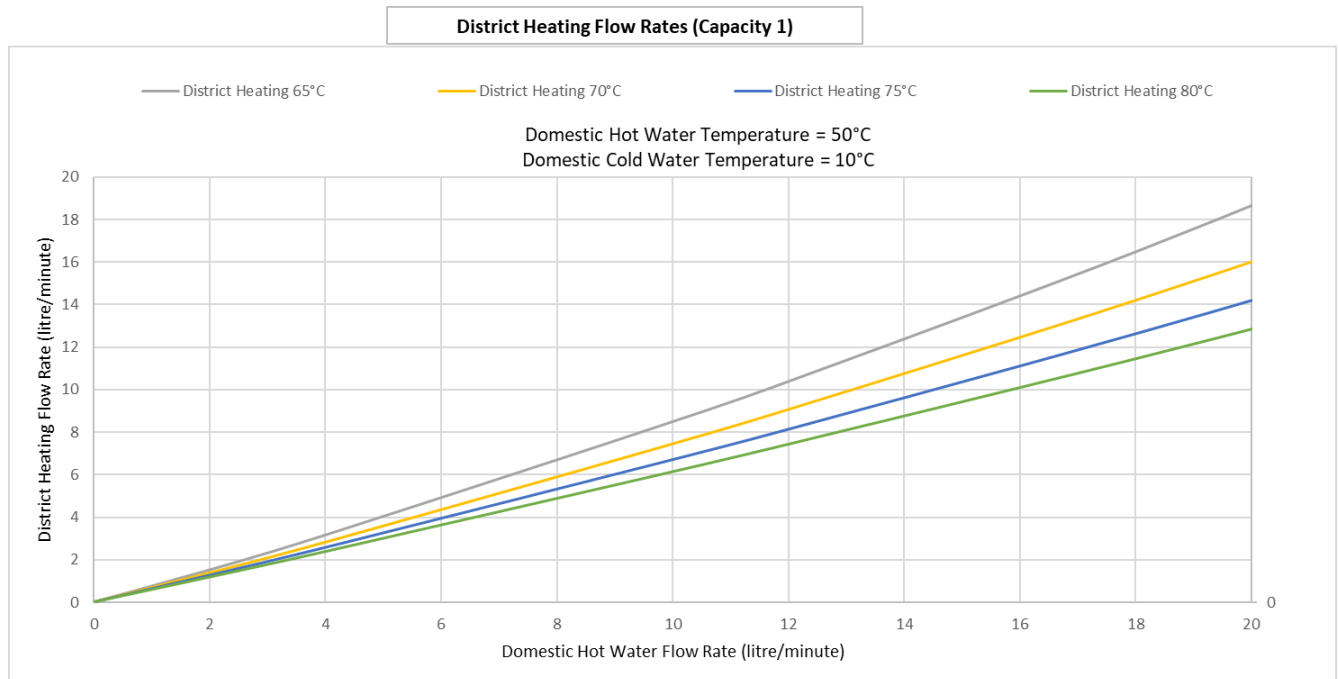
** Unit weight may vary depending on the optional components built into the unit.



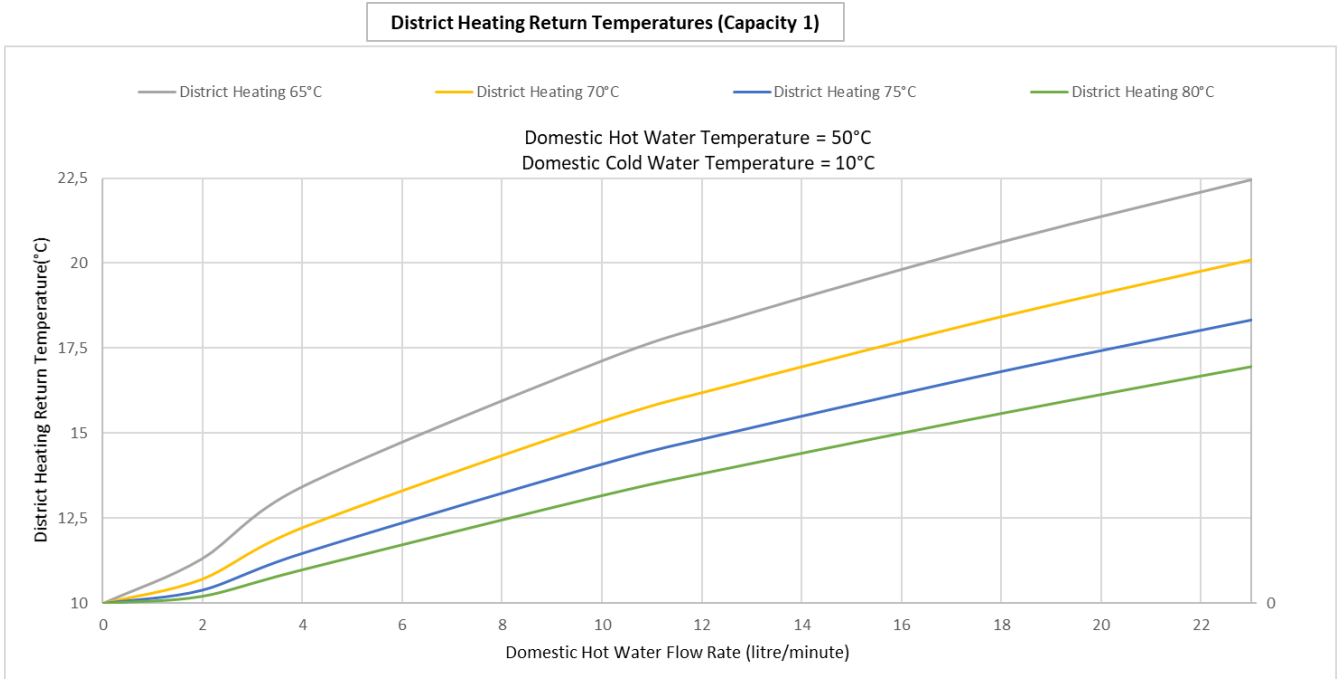
Indirect Smart Hexa

4. Performance Curves

4.1 District Heating Flow Rates (Capacity 1)



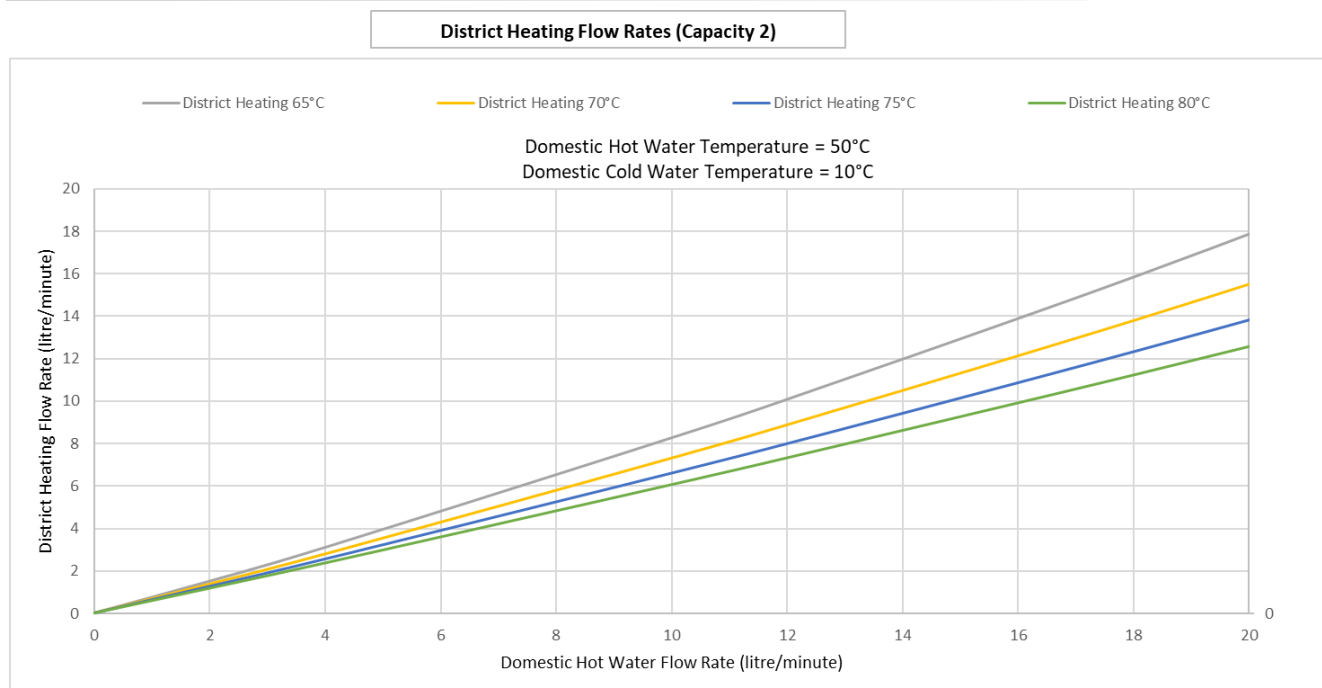
4.2 District Heating Return Temperatures (Capacity 1)



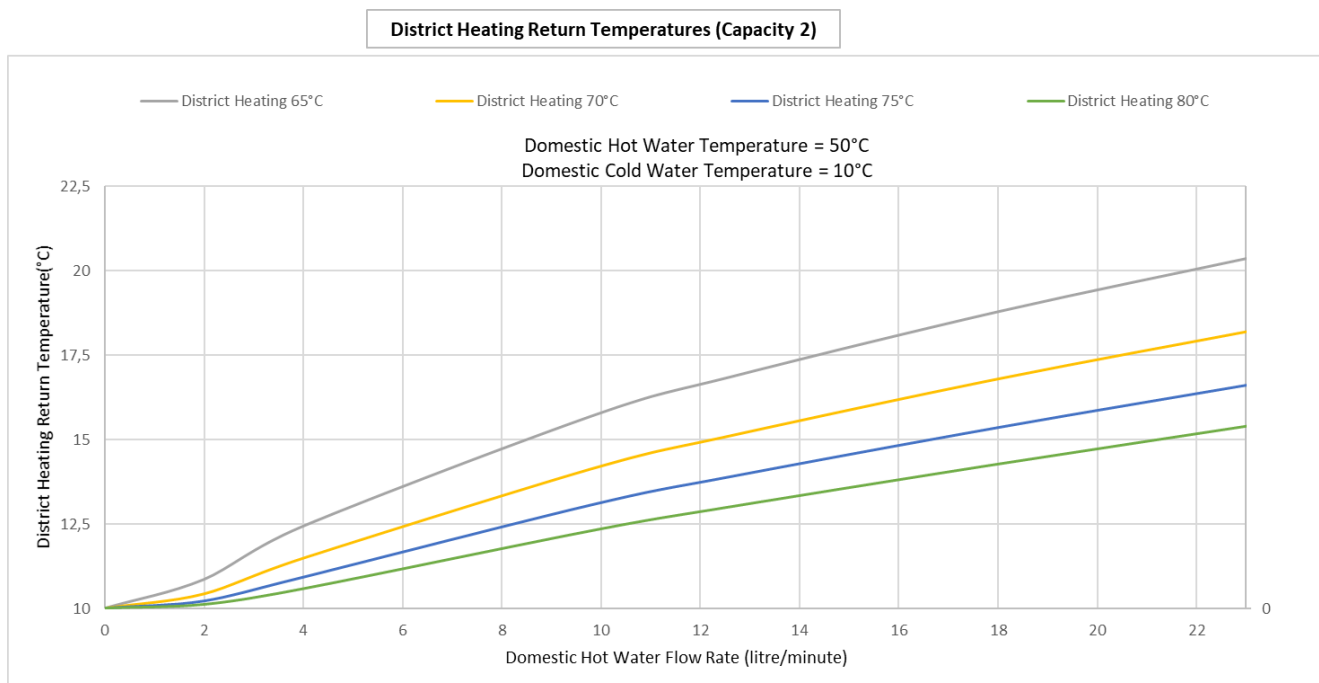
Indirect Smart Hexa

4. Performance Curves

4.3 District Heating Flow Rates (Capacity 2)



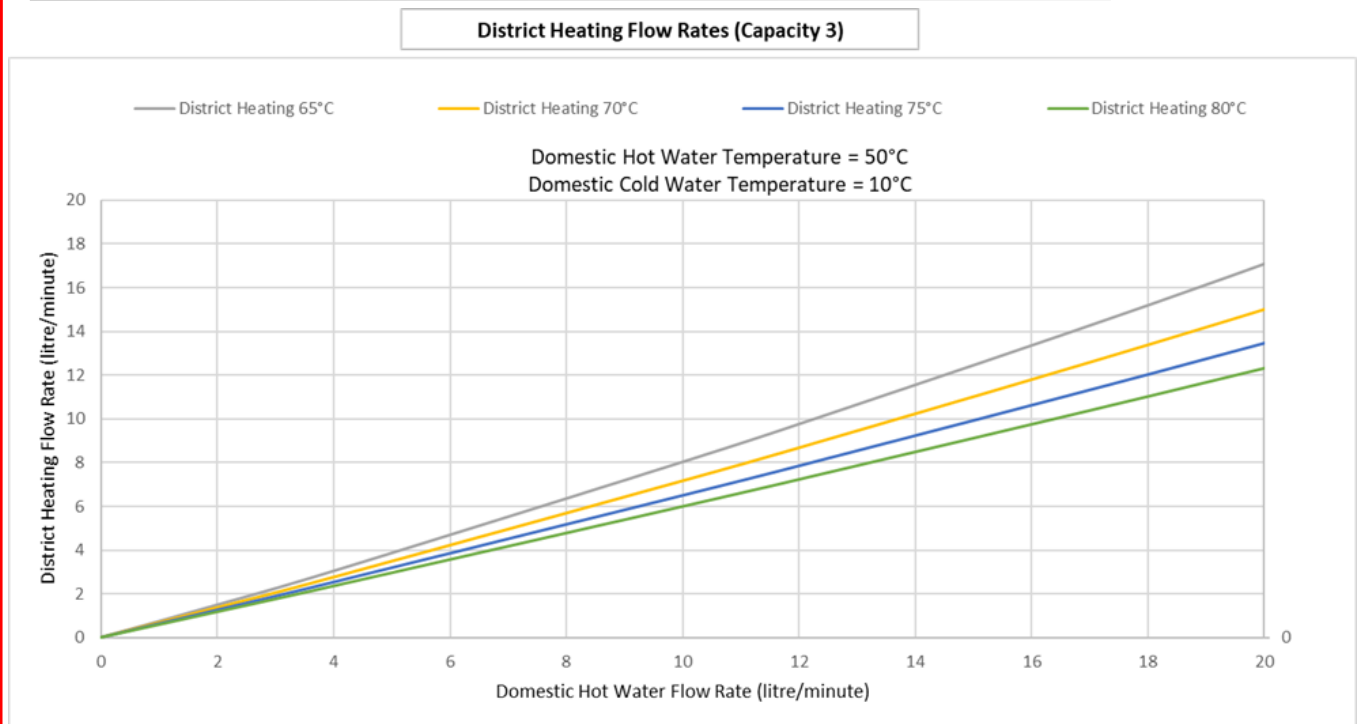
4.4 District Heating Return Temperatures (Capacity 2)



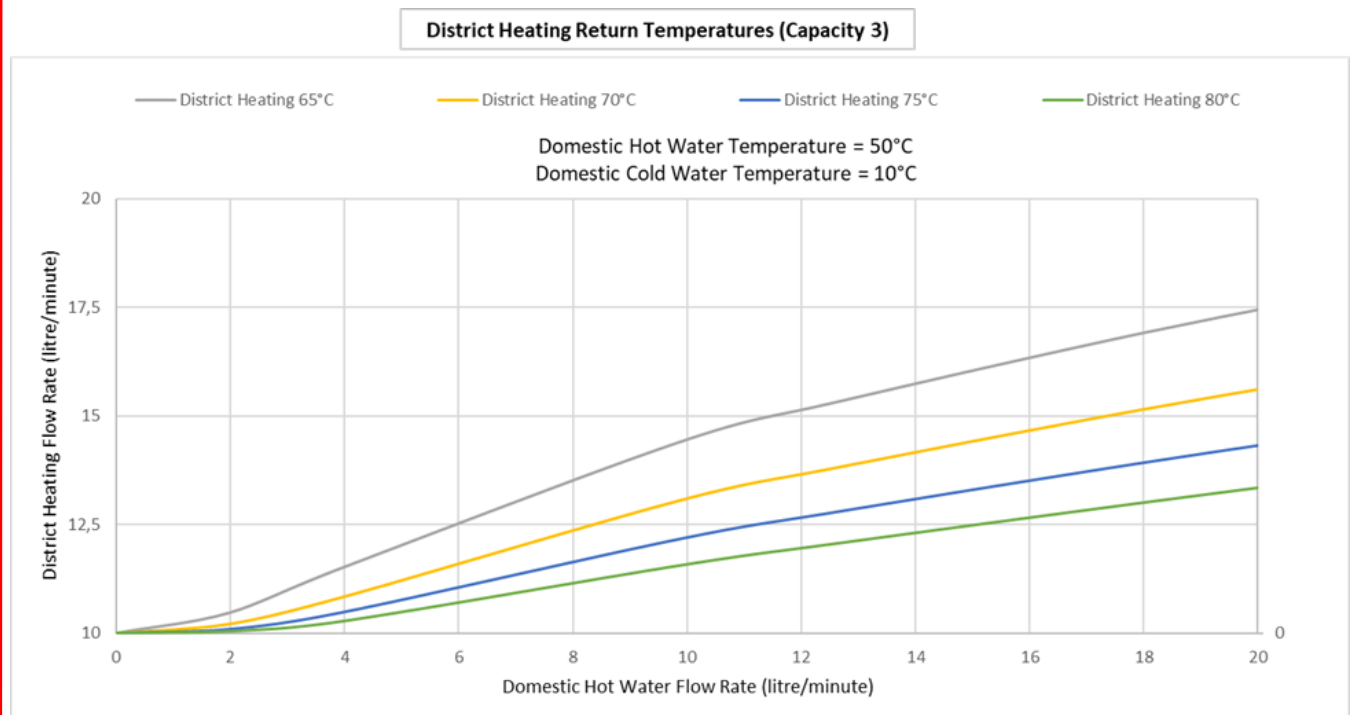
Indirect Smart Hexa

4. Performance Curves

4.5 District Heating Flow Rates (Capacity 3)



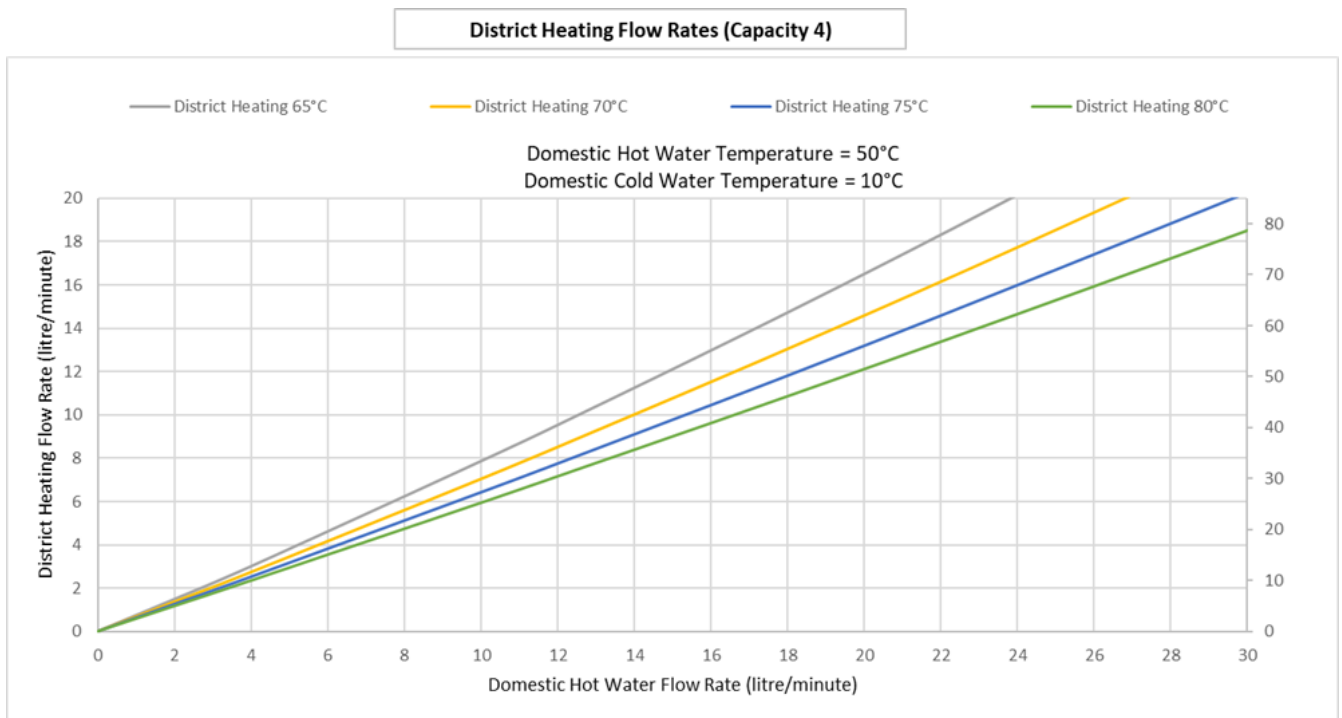
4.6 District Heating Return Temperatures (Capacity 3)



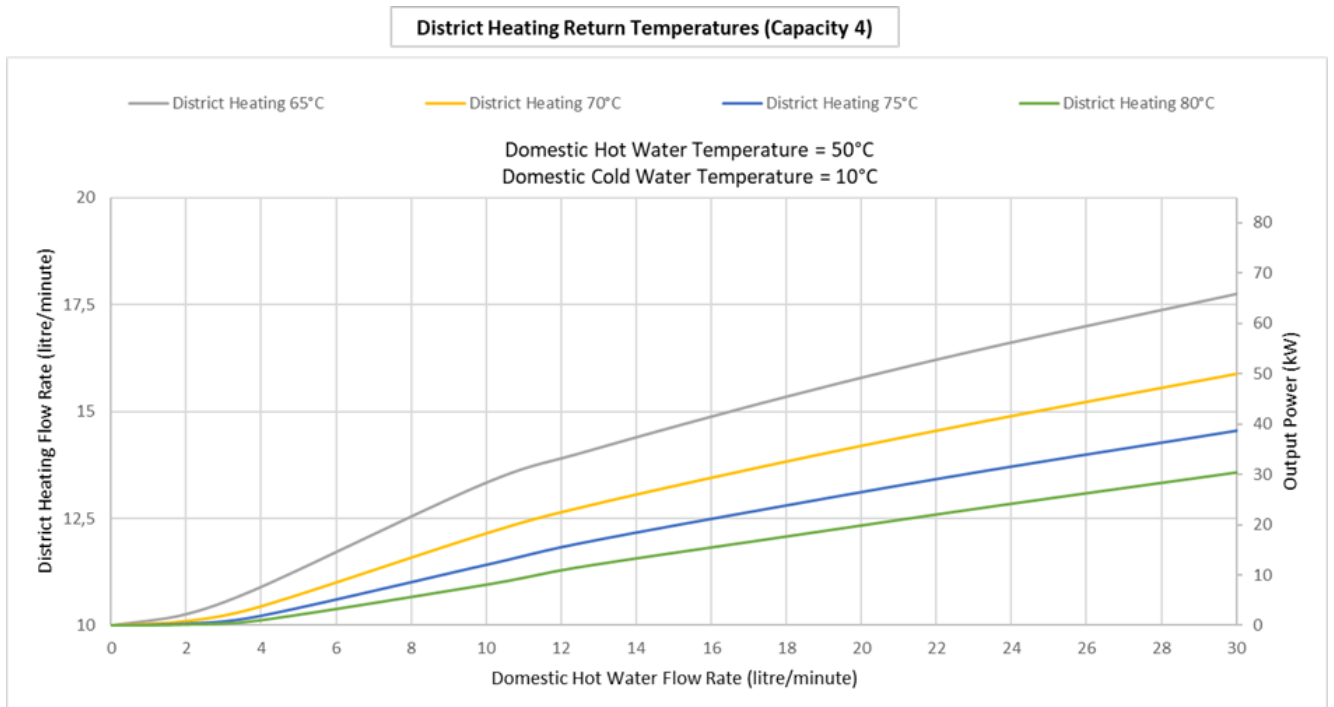
Indirect Smart Hexa

4. Performance Curves

4.7 District Heating Flow Rates (Capacity 4)



4.8 District Heating Return Temperatures (Capacity 4)



★ Indirect Smart Hexa

5. Performance Table

5.1 Capacity 1

HIU Performance of DHW Capacity 1					
Hot Water Output	Primary Flow °C	Capacity (kW)	DHW Flow Rate (l/m)	Primary Return °C	Primary Flow Rate (l/min)
10-50°C	55	5,5	1,983	15,36	2,021
		11	3,966	19,29	4,487
		27,5	9,915	24,54	13,15
		35	12,62	26,25	17,73
	60	5,5	1,983	12,53	1,692
		11	3,966	15,47	3,608
		27,5	9,915	19,78	9,986
		35	12,62	21,27	13,2
	65	5,5	1,983	11,31	1,5
		11	3,966	13,41	3,122
		27,5	9,915	17,08	8,404
		35	12,62	18,38	10,99
	70	5,5	1,983	10,71	1,362
		11	3,966	12,21	2,795
		27,5	9,915	15,31	7,384
		35	12,62	16,43	9,594
	75	5,5	1,983	10,39	1,254
		11	3,966	11,46	2,55
		27,5	9,915	14,06	6,646
		35	12,62	15,04	8,596
	80	5,5	1,983	10,21	1,164
		11	3,966	10,98	2,354
		27,5	9,915	13,14	6,076
		35	12,62	14	7,833

Indirect Smart Hexa

5. Performance Table

5.2 Capacity 2

HIU Performance of DHW Capacity 2					
Hot Water Output	Primary Flow °C	Capacity (kW)	DHW Flow Rate (l/m)	Primary Return °C	Primary Flow Rate (l/min)
10-50°C	55	5,5	1,983	14,18	1,963
		11	3,966	17,75	4,301
		27,5	9,915	22,72	12,41
		35	12,62	24,27	16,59
		50	18,03	26,83	25,86
	60	5,5	1,983	11,8	1,667
		11	3,966	14,13	3,503
		27,5	9,915	18,23	9,616
		35	12,62	19,54	12,63
		50	18,03	21,77	19,1
	65	5,5	1,983	10,85	1,488
		11	3,966	12,42	3,063
		27,5	9,915	15,76	8,178
		35	12,62	16,87	10,65
		50	18,03	18,81	15,85
	70	5,5	1,983	10,42	1,356
		11	3,966	11,47	2,76
		27,5	9,915	14,18	7,235
		35	12,62	15,12	9,365
		50	18,03	16,81	13,8
	75	5,5	1,983	10,21	1,25
		11	3,966	10,91	2,528
		27,5	9,915	13,1	6,543
		35	12,62	13,9	8,435
		50	18,03	15,36	12,35
	80	5,5	1,983	10,11	1,163
		11	3,966	10,57	2,34
		27,5	9,915	12,33	6,003
		35	12,62	13,01	7,717
		50	18,03	14,28	11,24

Indirect Smart Hexa

5. Performance Table

5.3 Capacity 3

HIU Performance of DHW Capacity 3					
Hot Water Output	Primary Flow °C	Capacity (kW)	DHW Flow Rate (l/m)	Primary Return °C	Primary Flow Rate (l/min)
10-50°C	55	5,5	1,983	13,02	1,908
		11	3,966	15,83	4,091
		27,5	9,915	20,72	11,69
		35	12,62	22,08	15,49
		50	18,03	24,35	23,76
		65	23,43	26,22	32,89
	60	5,5	1,983	11,14	1,644
		11	3,966	12,83	3,406
		27,5	9,915	16,6	9,255
		35	12,62	17,7	12,08
		50	18,03	19,59	18,07
		65	23,43	21,22	24,48
	65	5,5	1,983	10,47	1,477
		11	3,966	11,51	3,012
		27,5	9,915	14,42	7,961
		35	12,62	15,32	10,32
		50	18,03	16,92	15,23
		65	23,43	18,32	20,39
	70	5,5	1,983	10,21	1,351
		11	3,966	10,83	2,73
		27,5	9,915	13,07	7,093
		35	12,62	13,81	9,147
		50	18,03	15,16	13,39
		65	23,43	16,37	17,8
	75	5,5	1,983	10,09	1,248
		11	3,966	10,48	2,511
		27,5	9,915	12,18	6,447
		35	12,62	12,79	8,285
		50	18,03	13,93	12,06
		65	23,43	14,98	15,95
	80	5,5	1,983	10,04	1,161
		11	3,966	10,27	2,33
		27,5	9,915	11,57	5,936
		35	12,62	12,07	7,61
		50	18,03	13,02	11,03
		65	23,43	13,94	14,53

Indirect Smart Hexa

5. Performance Table

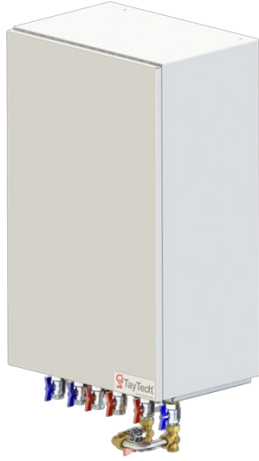
5.4 Capacity 4

HIU Performance of DHW Capacity 4					
Hot Water Output	Primary Flow °C	Capacity (kW)	DHW Flow Rate (l/m)	Primary Return °C	Primary Flow Rate (l/min)
10-50°C	55	5,5	1,983	12,12	1,869
		11	3,966	14,29	3,936
		27,5	9,915	18,99	11,12
		35	12,62	20,17	14,64
		50	18,03	22,16	22,18
		65	23,43	23,8	30,35
		85	30,65	25,67	42,21
	60	5,5	1,983	10,7	1,629
		11	3,966	11,87	3,338
		27,5	9,915	15,24	8,974
		35	12,62	16,16	11,66
		50	18,03	17,76	17,29
		65	23,43	19,12	23,22
		85	30,65	20,72	31,61
	65	5,5	1,983	10,26	1,471
		11	3,966	10,89	2,977
		27,5	9,915	13,31	7,791
		35	12,62	14,06	10,06
		50	18,03	15,36	14,75
		65	23,43	16,51	19,63
		85	30,65	17,88	26,42
	70	5,5	1,983	10,1	1,348
		11	3,966	10,44	2,712
		27,5	9,915	12,13	6,979
		35	12,62	12,78	8,983
		50	18,03	13,84	13,07
		65	23,43	14,8	17,29
		85	30,65	15,99	23,11
	75	5,5	1,983	10,04	1,247
		11	3,966	10,22	2,501
		27,5	9,915	11,4	6,369
		35	12,62	11,94	8,175
		50	18,03	12,81	11,84
		65	23,43	13,63	15,6
		85	30,65	14,64	20,74
	80	5,5	1,983	10,02	1,161
		11	3,966	10,12	2,325
		27,5	9,915	10,94	5,882
		35	12,62	11,38	7,535
		50	18,03	12,08	10,87
		65	23,43	12,77	14,28
		85	30,65	13,65	18,92

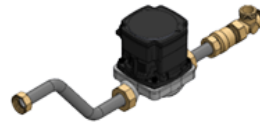
Indirect Smart Hexa

6. Options

Decorative Steel Cover



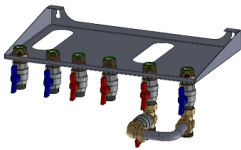
DHW Re-Circulation Kit



Junction Box



Pre-Plumbing Brackets
for Easy Assembly



Water Hammer Arrestor



Heat Meter



Flushing By-Pass Kit

